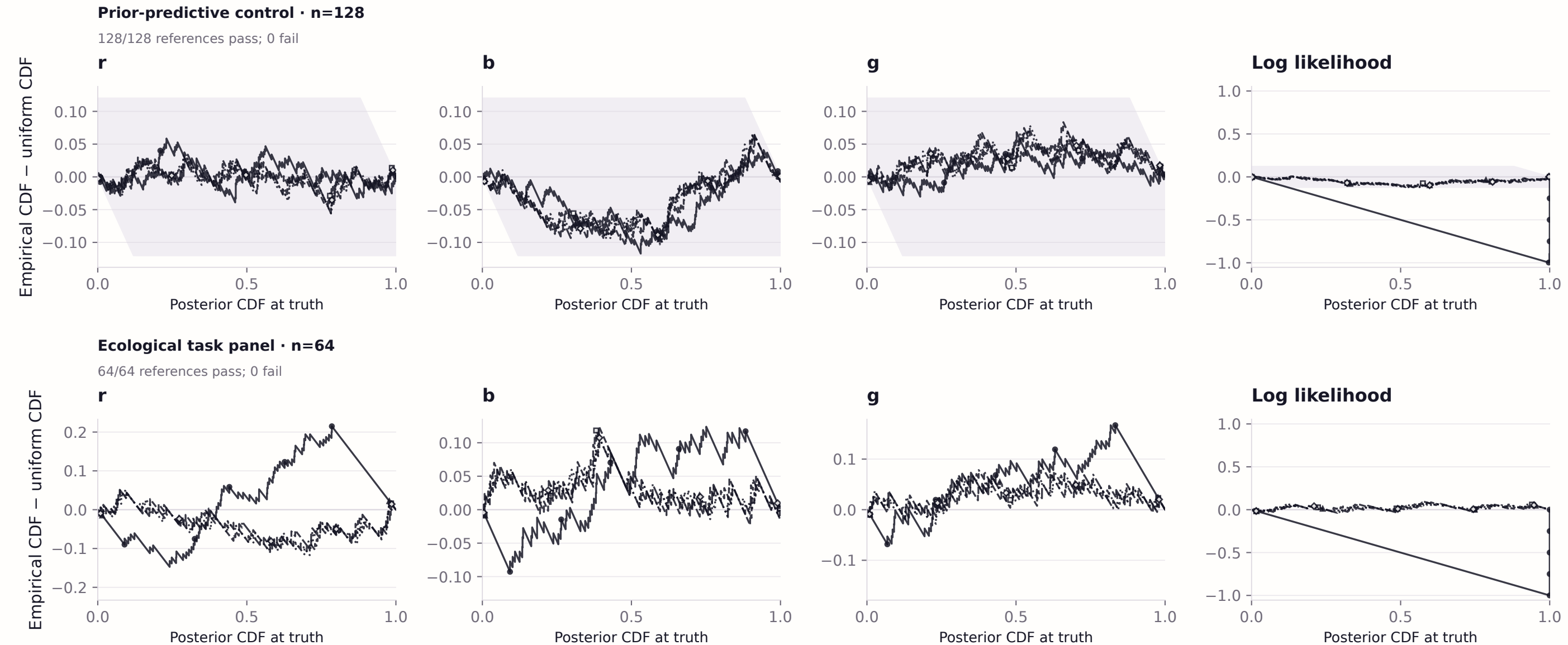


Posterior CDF diagnostics separate calibration from task behavior

CDF at simulator truth · one value per independent case · all cases retained, including failed reference outcomes



Top gray band: 95% DKW simultaneous bound for one independent-case ECDF under uniform calibration; not a familywise test across panels or methods.

Bottom: descriptive task diagnostics; ecological feature generation is outside the conditional likelihood, so uniformity is not the asserted null.

A frozen prior can pass marginal prior-predictive checks without learning. The data-dependent likelihood statistic provides a complementary check.

—●— Frozen prior -□- Published SMC · 1,024 / 4 ...▲... Higher compute · 4,096 / 8 —◇— Batch MCMC reference